# Circles

## Starter



1. Complete the square: write each expression as $(x+p)^2+q$.

**1(a)** $x^2+10x+28$

1(a)

**1(b)** $x^2-6x+1$

1(b)

**1(c)** $x^2-12x$

1(c)

**1(d)** $x^2+7x$

1(d)

2. Find the line through each pair of points.

**2(a)** A$(0,-6)$, B$(4,3)$

2(a)

**2(b)** P$(7,-5)$, Q$(-9,3)$

2(b)

**2(c)** R$(-4,-2)$, T$(5,10)$

2(c)

3. Use the discriminant to determine the number of real solutions.

**3(a)** $x^2-7x+14=0$

3(a)

**3(b)** $x^2+11x+8=0$

3(b)

**3(c)** $4x^2+12x+9=0$

3(c)

**4** Find the line through $(3,-4)$ perpendicular to $6x-5y-1=0$.

4

## Chapter Overview

**Luisa · 3**

**1. Circle equations**

Find centres, radii and equations.

**2. Lines and circles**

Find intersections and identify tangency.

**3. Chords and tangents**

Use perpendicular bisectors and circle geometry.

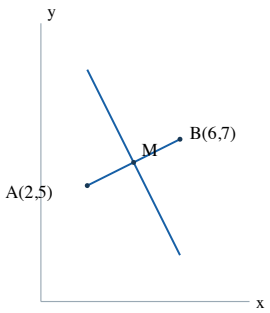
**4. Triangles in circles**

Find the circle through three vertices.

## Midpoints and Perpendicular Bisectors

**Luisa · 4**

A perpendicular bisector passes through the midpoint of a segment and meets it at a right angle.




**1** Find the midpoint of $A(2,5)$ and $B(6,7)$.

1

**2** Find the gradient of the perpendicular bisector.

2

**3** Find its equation.

3

The perpendicular bisector of a chord passes through the centre of its circle.

## Test Your Understanding — Luisa · 5

**a** Find the perpendicular bisector of A(4,7), B(10,17).

a

**b** Find a general perpendicular-bisector equation for distinct points $A(x_1, y_1), B(x_2, y_2)$.

b

**c** AB is a diameter with centre $(5,-4)$. A$=(1,-2)$. Find B.

c

Derive the bisector from its midpoint and perpendicular direction; there is no need to memorise the general expression.

## Exercise 6A - Y1 Page 115 — Luisa · 6

**6** JK is a diameter with $J(\frac{3}{4}, \frac{4}{3})$, $K(-\frac{1}{2}, 2)$. The centre lies on $y = 8x + b$. Find b.

6

**7** AB is a diameter with A$(0,-2)$, B$(6,-5)$. Show its centre lies on x$-$2y$-$10$=$0.

7

**2** F$(-2,5)$, G$(2,9)$ are diameter endpoints. Find the line through the centre perpendicular to FG.

2

**3** J$(0,-3)$, K$(4,-5)$ are diameter endpoints. Find the perpendicular line through the centre in integer standard form.

3

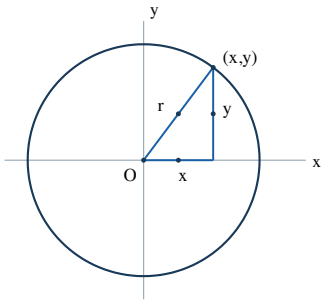


**5** X$=(7,-2)$, Y$=(4,q)$. Their perpendicular bisector is $y=4x+b$. Find q and b.

5

## Equation of a circle **Luisa · 9**

A point $(x, y)$ is on a circle centred at the origin with radius r. Draw a right-angled triangle from the origin to the point.

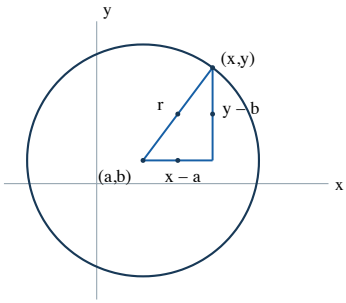



**Equation** What equation must x and y satisfy?

Equation

## Equation of a circle **Luisa · 10**

The centre moves to $(a, b)$. The horizontal and vertical displacements to $(x, y)$ are $x - a$ and $y - b$.




**Equation** Use Pythagoras to find the equation.

Equation

## Quickfire Questions

**Luisa · 11**

Find the missing equation, or the centre and radius, for each circle.

**a** Centre

$(0, 0)$

Radius

5

a

**b** Centre

$(1, 2)$

Radius

6

b

**c**

$(x + 3)^2 + (y - 5)^2 = 1$

c

**d**

$(x + 5)^2 + (y - 2)^2 = 49$

d

**e**

$(x + 6)^2 + y^2 = 16$

e

**f**

$(x - 1)^2 + (y + 1)^2 = 3$

f

**g**

$(x + 2)^2 + (y - 3)^2 = 8$

g

## Finding the equation using points

**Luisa · 12**

AB is a diameter with $\mathrm{A}(5,8)$, $\mathrm{B}(-7,4)$. Find the circle equation.

**1** Find the centre.

1

**2** Find the squared radius.

2

**3** Write the equation.

3

## Test Your Understanding — Luisa · 13

A=(5,−1), B=(13,11). AB is a diameter of circle C.

**a** Find the midpoint of AB.

a

**b** Find an equation for C.

b

## Completing the square — Luisa · 14

Find the centre and radius of

$x^2 + y^2 - 6x + 2y - 6 = 0$

**1** Group the x and y terms, then complete both squares.

1

**2** Read the centre and radius from the standard form.

2

## Further Example

**Luisa · 15**

A circle has equation

$x^2 + y^2 - 20x - 16y + 139 = 0$

**a** Find its centre.

a

**b** Show its radius is 5.

b

## Example

**Luisa · 16**

Derive the centre and radius of

$x^2 + y^2 + ax + by + c = 0$

**1** Complete the square.

1

**2** Identify the centre and radius.

2

A real circle requires the expression under the square root to be positive. Zero gives a single point; a negative value gives no real points.

**5** PQ is a diameter, with P(5,6) and Q(−2,2). Find the circle equation.

5

**6** The point $(1,-3)$ lies on $(x-3)^2+(y+4)^2=r^2$. Find r.

6



Complete the square, then find each centre and radius.

**8(a)**

$x^2+y^2-4x-11=0$

8(a)

**9(a)**

$x^2+y^2-10x+4y-20=0$

9(a)

8(b) Read the centre and radius.

8(b)

9(b) Read the centre and radius.

9(b)

## Exercise 6C - Y1 Page 120 — Luisa · 19

**12** P(7,−14) is on $x^2 + y^2 + 6x - 14y = 483$. PQ is a diameter. Find Q.

12

**13** The circle $(x - k)^2 + y^2 = 41$ passes through (3,4). Find both values of k.

13

## Extension — Luisa · 20

**1** [MAT 2009 1B] Find the shortest distance from the origin to $x^2 + y^2 + 6x + 8y = 75$.

1

**2** [MAT 2016 1I] For positive a,b and $x^2 + y^2 \leq 1$, find the greatest possible value of ax+by.

2

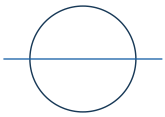
**3** [MAT 2007 1D] Which point on $(x - 5)^2 + (y - 4)^2 = 4$ is closest to $(x - 1)^2 + (y - 1)^2 = 1$?

3

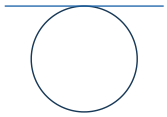
## Intersections of Lines and Circles



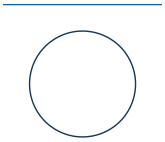
Solve a line and circle simultaneously by substitution. The resulting quadratic has two, one or no real roots: a secant, a tangent or no intersection.



Secant: two intersections



Tangent: one intersection



No intersection

**Example** Show that $y = x + 3$ never intersects $x^2 + y^2 = 1$.

Example

## Test Your Understanding



**1** Find where $y = x + 6$ meets $x^2 + (y - 3)^2 = 29$.

1

**2** Use algebra to find k when $y = x + k$ touches $x^2 + y^2 = 1$.

2

6. The line $x + y = 11$ meets the circle $x^2 + (y - 3)^2 = 32$.

**a** Show that there is only one intersection.

a

**b** Find the point of intersection.

b



7. The line $y = 2x - 2$ meets $(x - 2)^2 + (y - 2)^2 = 20$ at A and B.

**7(a)** Find A and B.

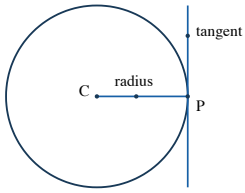
7(a)

**7(b)** Show that AB is a diameter.

7(b)

**11** Find the possible k for which $y = 4x - 1$ does not meet the circle $x^2 + 2x + y^2 = k$.

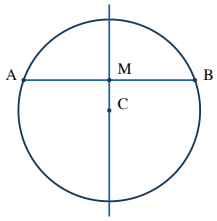
11




**Tangent and radius**

The tangent at a point on a circle is perpendicular to the radius to that point.

Find the radius gradient, take its negative reciprocal, then use the point to find the tangent equation.




**Chord and centre**

The perpendicular bisector of any chord passes through the centre.

Find the perpendicular bisectors of two chords. Their intersection is the circle centre.

## Examples



1. Circle C: $(x-3)^2+(y-7)^2=100$, with P(11,1).

1(a) Verify P is on C.

1(a)

1(b) Find the tangent at P.

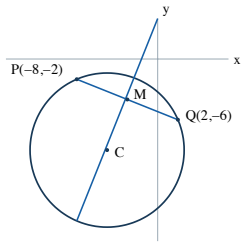
1(b)

2. Find both tangents of gradient $-3$ to $(x-4)^2+(y+4)^2=10$.

2

## Determining the Circle Centre — Luisa · 27

P(−8,−2) and Q(2,−6) lie on a circle with centre C. M is the midpoint of PQ. The line l passes through M and C.




**a** Find an equation of l.

a

**b(i)** Given that the y-coordinate of C is −9, show its x-coordinate is −5.

b(i)

**b(ii)** Find the circle equation.

b(ii)

## Test Your Understanding — Luisa · 28

**1** Find the tangent at P(6,9) to the circle with centre C(3,5). Use integer standard form.

1

**2** A circle passes through A(0,0), B(4,2). Its centre has x-coordinate −1. Find its equation.

2

6. The tangent to $(x+4)^2+(y-1)^2=242$ at $(7,-10)$ meets the y-axis at S and the x-axis at T.

**a** Find S and T.

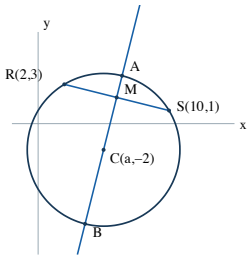
a

**b** Find the area of triangle OST, where O is the origin.

b



10. R(2,3), S(10,1) lie on a circle with centre C(a,−2). M is the midpoint of RS. The line l passes through M and C.




**a** Find l.

a

**b** Find a.

b

**c** Find the circle equation.

c

**d** Find where l meets the circle.

d

## Exercise 6E - Y1 Page 126-128 **Luisa · 31**

**1** [MAT 2012 1A] Which line is tangent to $x^2 + y^2 = 4$?

A: $x + y = 2$

B: $y = x - 2\sqrt{2}$

C: $x = \sqrt{2}$

D: $y = \sqrt{2} - x$

1

**2.** [**AEA 2006 Q4**]

$y = mx$ is tangent to $(x + 4)^2 + (y - 7)^2 = 13$.

(a) Show $3m^2 + 56m + 36 = 0$.

(b) Find the two contact points A and B.

(c) Tangents from $(4,-7)$ touch $x^2 + y^2 = 13$ at P and Q. Find P or Q.

Worked solution: next slide.

**3.** [**STEP 2005 Q6**]

A=(5,16), B=(−4,4), and AP=2BP.

(i) Show the locus of P is $(x + 7)^2 + y^2 = 100$.

(ii) C=(a,0), D=(b,0), QC=kQD, k>1. This gives the same locus.

Show $\dfrac{a + 7}{b + 7} = \dfrac{a^2 + 51}{b^2 + 51}$.

For a≠b, show $(a + 7)(b + 7) = 100$.

Worked solution: slide 33.

## Mark Scheme for Extension Question 2 **Luisa · 32**

[AEA 2006 Q4] Tangents from the origin to $(x + 4)^2 + (y - 7)^2 = 13$.

**a** Find the gradient condition.

a

**b** Find the contact points.

b

**c** Find a contact point for tangents from $(4,-7)$ to $x^2 + y^2 = 13$.

c

## Mark Scheme for Extension Question 3 — Luisa · 33

[STEP 2005 Q6] A=(5,16), B=(−4,4), AP=2BP.

**i** Find the locus.

i

**ii** Compare the locus QC=kQD with the same circle.

ii

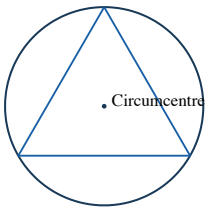
**Conclusion** Deduce the result when $a \neq b$.

Conclusion

## Triangles in Circles — Luisa · 34

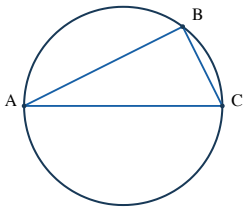
A triangle is inscribed in a circle when all three vertices lie on the circle.

The circle circumscribes the triangle. It is called the circumcircle; its centre is the circumcentre.



If $\angle ABC = 90°$, AC is the diameter of the circumcircle.

Conversely, if AC is a diameter and B is on the circle, $\angle ABC = 90°$. Thus AB is perpendicular to BC and $AB^2 + BC^2 = AC^2$.




To find the circumcentre of any triangle: find the perpendicular bisectors of two sides, then solve for their intersection.

## Example Luisa · 36

A=(−8,1), B=(4,5), C=(−4,9).

**a: method 1** Show AB is a diameter using Pythagoras.

a: method 1

**a: method 2** Show the two sides are perpendicular.

a: method 2

**b** Find the circle equation.

b

## Example



Find the circle through A(0,2), B(2,0) and C(8,18).

**1** Find two perpendicular bisectors.

1

**2** Find the centre and radius.

2

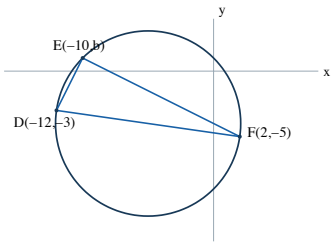
**3** Write the circle equation.

3

## Exercise 6F - Y1 Page 131-132



8. D(−12,−3), E(−10,b), F(2,−5) lie on a circle. Given $\angle DEF = 90°$ and b>0:




**a** Show b=1.

a

**b** Find the circle equation.

b

9. A circle has equation $x^2 + 2x + y^2 - 24y - 24 = 0$.

**a** Find its centre and radius.

a

**b** A$(-13,17)$ and B$(11,7)$ lie on the circle. Show AB is a diameter.

b

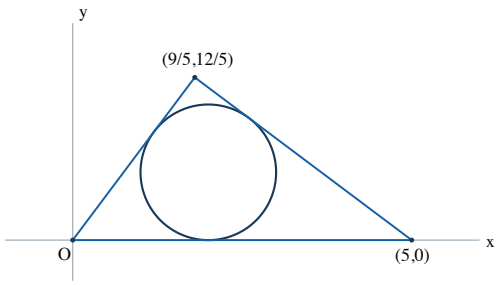
**c** C lies on the negative x-axis and angle ACB is a right angle. Find C.

c



[STEP] A circle touches all three sides of the triangle bounded by $y = 0$, $3y = 4x$ and $4y + 3x = 15$. Find its equation.

Given: the family $(x - 2t)^2 + (y - t)^2 = t^2$, t>0, touches the first two lines. The incircle lies entirely inside the triangle.




**1** Use the first two sides.

1

**2** Use the third side.

2

**3** Write the equation.

3